Seminar: The Local Langlands Conjecture for GL(2)

First Semester: Automorphic Side

Reference: C. J. Bushnell and G. Henniart, *The Local Langlands Conjecture for GL(2)*, Springer, 2006.

Note on time. Each talk is 90 minutes.

Lecture 1 — §1 and §§2.1–2.3 (pp. 8–17): Smooth Representations

Locally profinite groups and their compact open subgroups; smooth and admissible representations; representations of compact groups; fixed vectors V^K and exactness of $V \mapsto V^K$.

Main goal: Set up the basic language.

Description: Explain what a locally profinite group is, using F, $\mathfrak{o}$ and $\mathrm{GL}_2(F)$ as running examples. The key point is the meaning of "smooth": every vector is fixed by some compact open subgroup. Use characters of $F^\times$, locally constant functions on $\mathrm{GL}_2(F)$ as examples. End with the fact that $V \mapsto V^K$ is exact; we use it again in Lecture 4.

Lecture 2 — §§2.4–2.10 (pp. 17–24): Induction, Schur's Lemma and the Contragredient

Smooth induction Ind_H^G and compact induction c-Ind_H^G; Frobenius reciprocity; the countability hypothesis and Schur's Lemma; central characters; the contragredient (smooth dual) and admissibility.

Main results: Frobenius Reciprocity (2.4), Schur's Lemma (2.6).

Description: Define Ind and c-Ind and prove Frobenius reciprocity. For Schur's Lemma, the idea is: an irreducible smooth representation has countable dimension (because G/K is countable), while $\mathbb{C}$ is uncountable, so every G-endomorphism is a scalar. Deduce that irreducible representations have a central character. Finish with the contragredient, which is needed in Lecture 3.

Lecture 3 — §3 (pp. 25–32): Haar Measure and Duality

Haar measures, modular characters, and semi-invariant measures; dual of a compactly induced representation.

Main proof: Duality Theorem 3.5:

$$\left(\text{c-Ind}_H^G \sigma\right)^\vee \simeq \mathrm{Ind}_H^G \left(\delta_{H\backslash G} \otimes \check{\sigma}\right).$$

Description: Point out that on a locally profinite group, integrating a locally constant function with compact support is just a finite sum over cosets of a compact open subgroup. Explain the modular character and semi-invariant measures on $H\backslash G$, then prove the Duality Theorem 3.5.

Lecture 4 — §4 (pp. 33–42): The Hecke Algebra

The Hecke algebra $H(G)$ with convolution; idempotents e_K attached to compact open subgroups; the algebra $H(G, K)$ and its action on V^K; from smooth representations of G to modules over Hecke algebras.

Main proof: 4.3 Proposition: $V \mapsto V^K$ gives a bijection between irreducible smooth representations with $V^K \neq 0$ and simple $H(G,K)$-modules.

Description: Define $H(G)$ and $H(G,K)$ and show how they act on V and on V^K. The main message: to study representations with nonzero K-fixed vectors, it is enough to study modules over $H(G,K)$, which is a concrete algebra. Prove 4.3 carefully. The more general version in 4.4 and the material in 4.5 can be stated without proof.

Lecture 5 — §§5–7 (pp. 43–55): GL_2 over a Finite Field; Decompositions of $\mathrm{GL}_2(F)$

The subgroups B, N, T, Z of $\mathrm{GL}_2(k)$ and the Bruhat decomposition; irreducible representations of $\mathrm{GL}_2(k)$: those coming from induction from B, and the cuspidal ones; the cuspidal representations π_θ attached to regular characters θ of $l^\times$ (l/k quadratic) and their character table (6.4.1). Over F: Iwasawa and Cartan decompositions (7.2), the Iwahori subgroup (7.3), Haar measure on G and the Duality Theorem 7.7.

Main result: 6.4 Theorem: every irreducible cuspidal representation of $\mathrm{GL}_2(k)$ is π_θ for a regular character θ, and $\pi_{\theta_1} \cong \pi_{\theta_2}$ exactly when $\theta_2 = \theta_1$ or θ_1^q.

Description: Explain the basic split: an irreducible representation of $\mathrm{GL}_2(k)$ either contains the trivial character of N (then it comes from a representation induced from B) or it does not (then it is cuspidal). Present the character table (6.4.1) and how it gives the theorem; you may skip parts of the computation. For §5 and §7, state the decompositions with short proofs by row reduction, and state 7.7 without proof (it is used in Lecture 6). Suggested split: about 60 minutes on §§5–6, 30 minutes on §7.

Lecture 6 — §§8–9 (pp. 56–68): The Mirabolic Group and Principal Series

The mirabolic subgroup M and its representations; Jacquet modules; every irreducible non-cuspidal representation sits inside some $\mathrm{Ind}_B^G \chi$; reducibility of $\mathrm{Ind}_B^G \chi$; classification of the irreducible non-cuspidal representations.

Main proof: Irreducibility Criterion 9.6 for $\mathrm{Ind}_B^G(\chi_1 \otimes \chi_2)$.

Description: Start by saying why we look at the mirabolic group M: its smooth representations are easy to describe, and this description controls how $\mathrm{GL}_2(F)$-representations restrict to M. Then define the Jacquet module and the meaning of "cuspidal" via Jacquet modules. Prove the Irreducibility Criterion 9.6, and end with the list of irreducible non-cuspidal representations (9.10–9.11).

Lecture 7 — §§10–11 (pp. 69–84): Cuspidal Representations and Compact Induction

Matrix coefficients; cuspidal representations are exactly those with coefficients compactly supported modulo the centre; the Projectivity Theorem (10a); intertwining; compact induction from open subgroups that are compact modulo the centre; the level-zero example 11.5.

Main proof: 11.4 Theorem: if g intertwines ρ only when $g \in K$, then c-$\mathrm{Ind}_K^G \rho$ is irreducible and cuspidal.

Description: From §10, state the theorem that "cuspidal" can be read off from matrix coefficients, with at most a sketch, and state the Projectivity Theorem of §10a without proof. Spend most of the time on §11: define intertwining, prove 11.4, and work out 11.5 in detail.

Example 11.5 shows that level-zero cuspidal representations come from cuspidal representations of $\mathrm{GL}_2(k)$ (Lecture 5). Suggested split: 20 minutes on §§10–10a, 70 minutes on §11.

Lecture 8 — §§12–13 (pp. 86–99): Chain Orders and Fundamental Strata

Chain orders $\mathfrak{M}$ and $\mathfrak{I}$; filtrations of unit groups; strata $(\mathfrak{A}, n, a)$ and their characters ψ_a; the normalized level $\ell(\pi)$; fundamental strata; split and simple strata.

Main result: 12.9 Theorem: a stratum contained in π is fundamental if and only if its level equals $\ell(\pi)$; in particular π contains a fundamental stratum if and only if $\ell(\pi) > 0$. Also 13.3 Corollary: if $\ell(\pi) > 0$, then π contains either a split fundamental stratum or a simple stratum.

Description: Strata are a tool to measure how ramified a representation is. For GL_2 only two chain orders matter up to conjugacy, $\mathfrak{M} = M_2(\mathfrak{o})$ and the Iwahori order $\mathfrak{I}$, so work with these two explicitly. Explain how a stratum gives a character of a unit group, and what "fundamental" means. In §13, describe the different kinds of fundamental strata with examples, and keep the ring-theoretic proofs short.

Lecture 9 — §§14–15 (pp. 100–110): The Exhaustion Theorem and the Classification of Cuspidal Representations

Split strata only occur in non-cuspidal representations; the level-zero case; the Exhaustion Theorem; intertwining and conjugacy of simple strata (stated); the groups J_α and the representations $\text{c-Ind}_{J_\alpha}^G \Lambda$; cuspidal types; cuspidal inducing data.

Main results: Exhaustion Theorem 14.5 and the Induction Theorem 15.5: every irreducible cuspidal representation of $\mathrm{GL}_2(F)$ is $\text{c-Ind}_J^G \Lambda$ for a cuspidal type $(\mathfrak{A}, J, \Lambda)$, and the type is unique up to conjugacy.

Description: Present it in two steps. Step 1 (Exhaustion Theorem): a cuspidal representation, twisted to have minimal level, contains either a cuspidal representation of $\mathrm{GL}_2(k)$ (level zero) or a simple stratum. Step 2 (§15): from a simple stratum, build J_α and Λ; then c-Ind is irreducible and cuspidal (15.3, using 11.4), and π determines the type up to conjugacy (15.4). State the Intertwining and Conjugacy Theorems (15.1, 15.2) without proof; they are proved in Lecture 10. Finish with cuspidal inducing data (15.8), which are needed in Lectures 11 and 13. Suggested split: 30 minutes on §14, 60 minutes on §15.

Lecture 10 — §§16–17 (pp. 111–122): Proofs of the Intertwining Results; Iwahori-Fixed Vectors

Proofs of the Intertwining Theorem 15.1, the Conjugacy Theorem 15.2 and 15.6 Lemma; the Iwahori–Hecke algebra $H(G, \mathfrak{I})$; end of the proof of 14.3 Lemma; square-integrable representations and the Steinberg representation.

Main result: 17.5 Theorem: the Steinberg representation is square-integrable, and a non-cuspidal irreducible representation is square-integrable exactly when it is $\phi \cdot \mathrm{St}_G$ with $|\phi| = 1$.

Description: First half: explain the idea of the proof of the Intertwining Theorem 15.1, and only state what the other proofs in §16 give. Second half: introduce the explicit elements of $H(G, \mathfrak{I})$ used in §17 and use them to show that the Steinberg representation is square-integrable. Suggested split: 40 minutes on §16, 50 minutes on §17.

Lecture 11 — §§18–21 (pp. 123–133): Admissible Pairs and Tame Parametrization

Admissible pairs $(E/F, \chi)$: E/F a tamely ramified quadratic extension and χ a character of $E^\times$ satisfying the conditions of 18.2; construction of the cuspidal representation π_χ (level zero via §6, positive level via simple strata); the Tame Parametrization Theorem; tame intertwining properties used in its proof.

Main result: Tame Parametrization Theorem 20.2: $(E/F, \chi) \mapsto \pi_\chi$ is a bijection from admissible pairs to all irreducible cuspidal representations of $\mathrm{GL}_2(F)$ if $p \neq 2$, and to the unramified ones if $p = 2$.

Description: This talk turns the classification of Lecture 9 into an explicit list: for $p \neq 2$, cuspidal representations of $\mathrm{GL}_2(F)$ are labelled by characters of tamely ramified quadratic extensions. Define admissible pairs carefully, show how to build π_χ in the level-zero case and in one positive-level case, and state the properties (1)–(4) in 20.2. State 19.4 Proposition without proof (it is finished in Lecture 12), and only outline §21.

Lecture 12 — §§22–23 (pp. 134–146): A Group Extension and the Functional Equation for GL_1

A finite p-group extension, used to finish the proof of 19.4 Proposition; zeta integrals for characters of $F^\times$; the local functional equation, $L(\chi, s)$ and $\varepsilon(\chi, s, \psi)$; Gauss sums and the table of values of ε; the Stability Theorem 23.8.

Main results: the local functional equation for GL_1 and the formula for $\varepsilon(\chi, s, \psi)$ by Gauss sums (23.5).

Description: Spend at most 20 minutes on §22: state its result and show where it is used in 19.4. The main part is §23, the local part of Tate's thesis. Define the zeta integral, prove the functional equation, and compute ε for ramified and unramified characters. Finish with the Stability Theorem 23.8. The same pattern is repeated for GL_2 in Lecture 13.

Lecture 13 — §§24–25 (pp. 147–161): Functional Equation for GL_2 and Cuspidal Local Constants

Godement–Jacquet zeta integrals $\zeta(\Phi, f, s)$ with $\Phi \in C_c^\infty(M_2(F))$ and f a matrix coefficient of π; the functional equation, $L(\pi, s)$ and $\varepsilon(\pi, s, \psi)$; proof for cuspidal π; for cuspidal π, $L(\pi, s) = 1$ and $\varepsilon(\pi, s, \psi)$ is given by a Gauss sum of a cuspidal inducing datum; the level-zero case; stability.

Main results: 24.2 Theorems 1 and 2 (functional equation) and 25.2 Theorem (the Gauss sum formula for $\varepsilon(\pi, s, \psi)$).

Description: This is Lecture 12 again in dimension two: replace F by $M_2(F)$ and characters by matrix coefficients. Note that L and ε here are defined using $\mathrm{GL}_2(F)$ alone; the comparison with the Weil group side comes in the second semester. Prove the functional equation for cuspidal π in outline only. Then compute $\varepsilon(\pi, s, \psi)$ for cuspidal π from its cuspidal inducing datum (Lecture 9), and do the level-zero case explicitly with the character table from Lecture 5. Suggested split: 45 minutes on each section.

Lecture 14 — §§26–27 (pp. 162–178): Non-Cuspidal Local Factors and the Converse Theorem

L and ε for the composition factors of $\mathrm{Ind}_B^G \chi$, including the Steinberg representation; the Converse Theorem: if $L(\chi\pi_1, s) = L(\chi\pi_2, s)$ and $\varepsilon(\chi\pi_1, s, \psi) = \varepsilon(\chi\pi_2, s, \psi)$ for all characters χ of $F^\times$, then $\pi_1 \cong \pi_2$.

Treatment of §27: Introduction only. State the theorem, explain what information it uses and why it is a uniqueness statement, and outline the structure of the proof. No full proof is planned.

Description: First give the formulas for L and ε of non-cuspidal representations in terms of χ_1, χ_2 (26.1), without the full computation. Then state the Converse Theorem and explain the structure of the proof: by 27.2, π is cuspidal if and only if $L(\phi\pi, s) = 1$ for all ϕ; the principal series case is then easy (27.3); the cuspidal case is the hard part and should only be described in words. Suggested split: 40 minutes on §26, 30 minutes on §27, 20 minutes recap.

Lecture time:

- Talk 1: October 30, 10h-12h, WSC-S-U-3.01
- Talk 2: November 6, 10h-12h, WSC-S-U-3.01
- Talk 3: November 20, 10h-12h, WSC-S-U-3.01
- Talk 4: November 20, 16h-18h, WSC-S-U-3.01 (?)
- Talk 5: November 27, 10h-12h, WSC-S-U-3.01
- Talk 6: December 2 (Wednesday), 10h-12h, WSC-S-U-3.02
- Talk 7: December 11, 10h-12h, WSC-S-U-3.01
- Talk 8: December 11, 16h-18h, WSC-S-U-3.01 (?)
- Talk 9: December 18, 10h-12h, WSC-S-U-3.01
- Talk 10: January 8, 10h-12h, WSC-S-U-3.01
- Talk 11: January 15, 10h-12h, WSC-S-U-3.01
- Talk 12: January 22, 10h-12h, WSC-S-U-3.01
- Talk 13: January 29, 10h-12h, WSC-S-U-3.01
- Talk 14: February 5, 10h-12h, WSC-S-U-3.01